REQ2 Design Rationale

Sterilisation Box

We did not want to check if something is a Sterilisation Box in the code. So we made a value called STERILISING in the Abilities list. The Sterilisation Box uses this value when it is created. Because the Actor checks all the items it is carrying before checking itself any actor with the box will automatically have the STERILISING ability without needing code. Other parts of the code like the Consume Apple Action and the Drink From Puddle Action only need to ask the actor if it has this ability. They do not need to know about the Sterilisation Box.

Poison Status

The Game Entity in the engine has a list of Status objects. Every turn it calls a function on each one. Then removes any that are not active. The Poison Status works with this system. Because it is an object for each poison effect multiple effects can be on the same actor at the same time.

Fire Ground

When the Lantern sets a tile on fire it replaces the ground with a Fire Ground. The Fire Ground remembers what the old ground was. After a turns the Fire Ground changes back to the old ground. This way we do not need a list of what used to be on each tile. The Fire Ground remembers for us.

Lantern

The Lantern does not have a use action. It does everything it needs to do every turn without the player needing to do anything. This keeps the players menu simple. Makes the fire start automatically like it is supposed to.

Puddle

The Puddle checks if the actor is standing on it or next to it. If the actor is standing on it the Puddle offers the drink action. We do not need to calculate the distance. The engine tells us this information.

Hole

The map already had a character, for the Hole so we had to make a Hole class to avoid an error. We made the class simple so the map would load correctly. We will add more to the Hole class later when we need to.